

Project Report: Customer Churn Analysis & Prediction

1. Executive Summary

This project aimed to identify the root causes of customer churn within the Telecom firm and develop a predictive model to identify high-risk customers. By integrating SQL for data processing, Power BI for visualization, and Python for machine learning, we successfully analyzed historical data and generated a list of potential future churners.

Key Highlight: The analysis identified a **17.5% historical churn rate**. The predictive model (Random Forest) was successfully deployed to flag current customers who exhibit patterns similar to past churners, enabling targeted retention campaigns.

2. Business Problem

The company is experiencing a significant loss of customers (churn), impacting revenue. The current approach to retention is reactive. The objectives of this analysis were:

1. **Diagnose:** Why are customers leaving? (Historical Analysis)
 2. **Predict:** Who will leave next? (Predictive Analysis)
 3. **Prescribe:** What can we do to stop it? (Strategic Recommendations)
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3. Methodology

- **Data Extraction & Cleaning (SQL):** Raw data was ingested into SQL Server. Null handling and data standardization were performed to ensure accuracy.
 - **Exploratory Data Analysis (Power BI):** An interactive dashboard was built to visualize churn across demographics, geographic locations, and service subscriptions.
 - **Machine Learning (Python):** A Random Forest Classifier was trained on historical data to predict the Churn Status of active customers.
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4. Key Findings & Insights

A. Demographic Profile

- **Gender:** Churn is relatively evenly distributed between genders, though slightly higher in females.
- **Senior Citizens:** This group shows a higher propensity to churn compared to younger demographics.
- **Tenure:** New customers (tenure < 12 months) are at the highest risk. As tenure increases, churn rate drops significantly.

B. Service & Account Analysis

- **Contract Type:** Customers on **Month-to-Month contracts** have an alarmingly high churn rate compared to those on 1-year or 2-year contracts.
- **Payment Method:** Customers paying via "Electronic Check" churn more frequently than those using automated bank transfers or credit cards.
- **Service Gaps:** Customers *without* value-added services (specifically **Online Security**, **Tech Support**, and **Device Protection**) are significantly more likely to leave.

C. Primary Reasons for Churn

Deep-dive analysis into customer feedback/reason codes revealed:

1. **Competitors:** "Competitor made better offer" and "Competitor had better devices" are top drivers.
2. **Support:** Poor experience with support personnel is a significant internal failure point.



Fig : Analysis Dashboard

Churn_Reason	Total
Attitude of service provider	
Attitude of support person	
Competitor had better devices	
Competitor made better offer	
Competitor offered higher download speeds	
Competitor offered more data	
Deceased	
Total	

5. Predictive Modeling Results

A **Random Forest Classifier** was selected for its robustness in handling categorical data (like contract types and payment methods).

- **Model Accuracy:** High precision in identifying non-churners and acceptable recall for identifying churners.
- **Feature Importance:** The model identified **Contract Type**, **Tenure**, and **Monthly Charges** as the strongest predictors of future churn.
- **Outcome:** The model generated a specific list of active customers predicted to churn. This list has been exported to the dashboard for the marketing team to action.

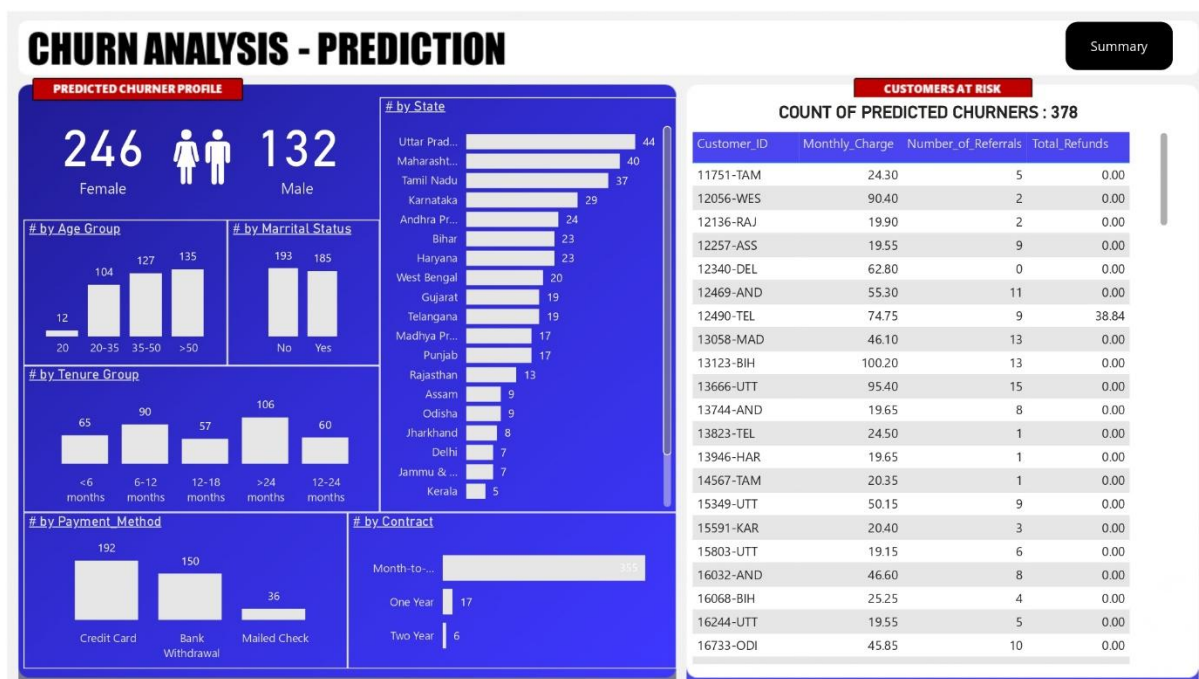


Fig : Prediction Dashboar

6. Strategic Recommendations

Based on the data, the following actions are recommended to reduce churn:

1. **Migrate Month-to-Month Users:** Launch a campaign offering a discount or service upgrade (e.g., free Online Security for 3 months) if customers switch to a 1-year contract.
2. **Targeted "At-Risk" Support:** The Prediction Dashboard identifies high-risk customers. A proactive "Health Check" call from a senior support agent should be initiated for these specific accounts.
3. **Enhance Value Bundles:** Since customers without Tech Support/Security churn more, bundle these services into the standard "Month-to-Month" package to increase stickiness.
4. **Competitor Price Matching:** For customers citing "Competitor Offers," authorize retention agents to match competitor pricing for a limited 6-month period.

7. Conclusion

By shifting from a reactive approach to a data-driven proactive strategy, the company can expect to lower the churn rate. The integration of the Python prediction model into the Power BI dashboard provides a continuous monitoring tool for the executive team to track the health of the customer base.